CHAPTER 11

Sequential Circuits, Latches & Flip-Flops

Memory · SR, D latches · SR, JK, D, T flip-flops · race-around · master-slave · setup, hold, clock-to-Q

QUICK TRICK

This is a big chapter, so learn these four lines first: SR: Set = 1, Reset = 0, 00 = hold, 11 = illegal. JK: 00 hold, 01 reset, 10 set, **11 toggle**. D: Q follows D. T: T = 1 toggles.

11.1 Combinational vs sequential

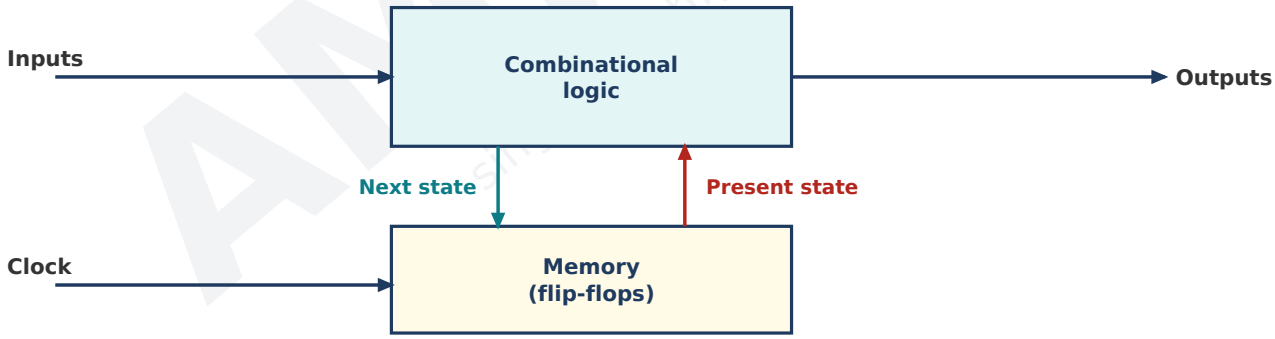
● **LEVEL 1 · BEGINNER**

Combinational circuit: the output depends only on the **present inputs**. It has no memory. (Adders, MUX, decoders.)

Sequential circuit: the output depends on the **present inputs and on what happened before**. It has **memory**.

Why do we need memory? A computer must remember numbers, counts and the step it is in. Without memory there would be no registers, no counters and no RAM.

Analogy. A calculator with a memory button. Pressing the same key can give different results, because it remembers earlier keys.



Word	Simple meaning
Memory element	A circuit that stores 1 bit (a latch or a flip-flop).
Feedback	The output is sent back to the input. This is how a circuit remembers.
Clock	A regular train of pulses (0, 1, 0, 1…). It tells the circuit **when** to change.
State	What the memory holds now (for example Q = 1).
Present state / Next state	State now / state after the next clock pulse.
Synchronous	All changes happen together at the clock.
Asynchronous	Changes can happen at any time (no common clock).

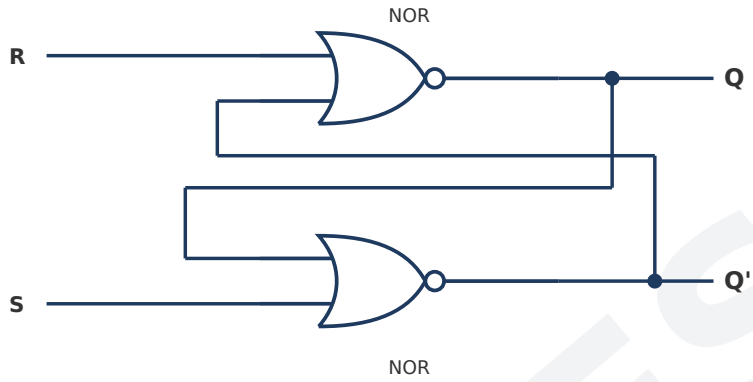
EASY MEMORY
Combinational = **no memory**. Sequential = **memory + feedback** (and usually a clock).

11.2 Latches

● **LEVEL 1 · BEGINNER**

What is a latch? The simplest 1-bit memory. It is built from two gates that **feed each other** (cross-coupled). A latch is **level-triggered**: it can change while its enable (or its inputs) are at a certain level.

SR latch with NOR gates



S = **Set** (make Q = 1). R = **Reset** (make Q = 0). Q' is the opposite of Q.

S	R	Q (next)	Meaning
0	0	Q (no change)	**Hold** — remembers the old value
0	1	0	**Reset**
1	0	1	**Set**
1	1	Q = Q' = 0	**Invalid** (and unpredictable when S, R return to 0)

SR latch with NAND gates (inputs are active-LOW)

S'	R'	Q (next)	Meaning
1	1	Q (no change)	**Hold**
1	0	0	**Reset**
0	1	1	**Set**
0	0	Q = Q' = 1	**Invalid**

QUICK TRICK
Invalid-state trick. The invalid case is when **both inputs are "active" at once**. NOR latch: active = 1, so invalid at **S = R = 1**. NAND latch: active = 0, so invalid at **S' = R' = 0**. The idle (hold) value is 0 for NOR and 1 for NAND.

Gated SR latch and D latch

A **gated latch** has an **enable (EN)** input. When EN = 0 the latch ignores S and R. The **D latch** removes the invalid state by using only one input D (S = D, R = D'):

EN	D	Q (next)	Meaning
0	X	Q	Hold (latch is closed)
1	0	0	Transparent: Q follows D
1	1	1	Transparent: Q follows D

QUICK TRICK
D latch = "transparent window". While EN = 1, Q copies D (like a glass window). When EN = 0 the window is closed and Q keeps the last value.

Latch	Flip-flop
Level-triggered (acts the whole time EN = 1)	**Edge**-triggered (acts only at the clock edge)
Q can change many times while EN = 1	Q changes only once per clock pulse
Simpler and faster	Safer for counters and registers
Used inside flip-flops	Built from two latches (master–slave)

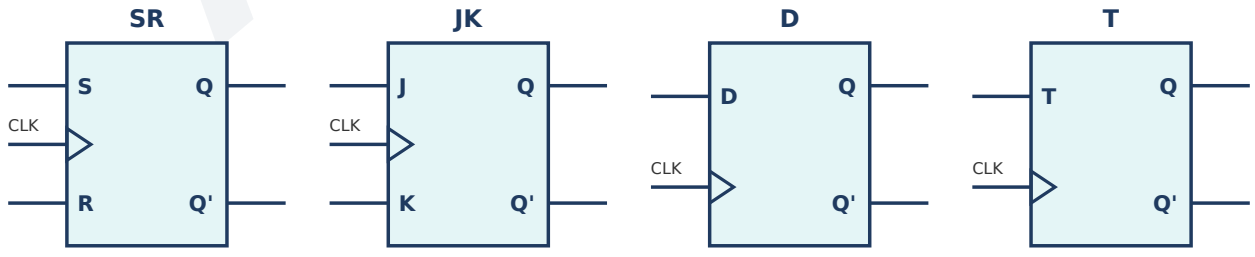
11.3 Flip-flops

● **LEVEL 2 · EXAM**

What is a flip-flop? A 1-bit memory that changes **only at a clock edge**. **Why?** It makes all the circuit change together in step, which avoids confusion.

Edge triggering. A **positive (rising) edge** is the moment the clock goes 0 → 1. A **negative (falling) edge** is 1 → 0. The flip-flop looks at its inputs only at that moment.

QUICK TRICK
Symbol trick: a small **triangle** at the clock pin = edge triggered. A triangle with a **bubble** = falling-edge triggered. Triangle only = rising-edge triggered. No triangle = latch or level triggered.



Notation: Q = present state, $\mathbf{Q^+}$ (or Q_{n+1}) = next state after the clock edge.

The four flip-flops — truth table, characteristic table, equation

Characteristic table = the next state Q^+ for every combination of the inputs *and* the present state Q. **Characteristic equation** = the same information written as a formula. X means invalid.

SR flip-flop

Q	S	R	Q+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

JK flip-flop

Q	J	K	Q+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

D flip-flop

Q	D	Q+
0	0	0
0	1	1
1	0	0
1	1	1

T flip-flop

Q	T	Q+
0	0	0
0	1	1
1	0	1
1	1	0

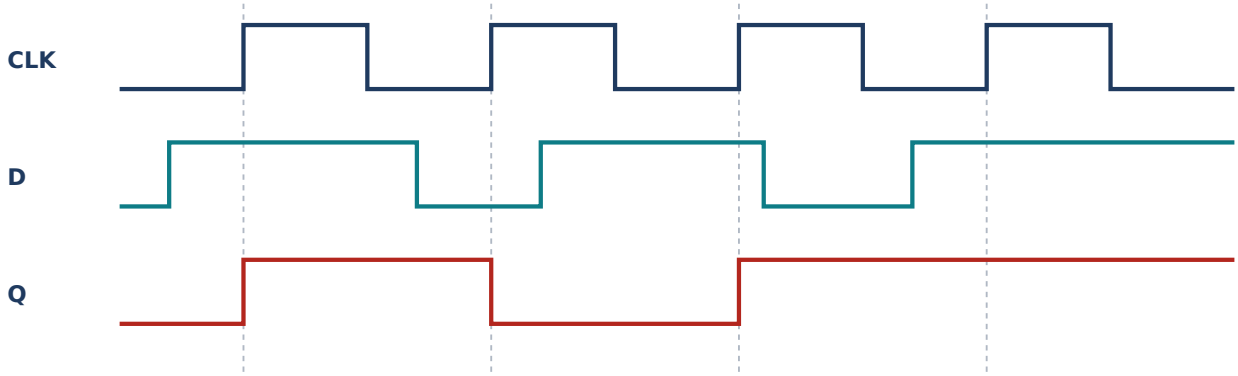
Flip-flop	Characteristic equation	What it does (in words)
SR	$Q^+ = S + R'Q$ (with SR = 0)	S sets, R resets, 00 holds, 11 is illegal
JK	$Q^+ = JQ' + K'Q$	00 hold, 01 reset, 10 set, 11 **toggle**
D	$Q^+ = D$	Q takes the value of D at the clock edge (a 1-clock delay)
T	$Q^+ = T \oplus Q = TQ' + T'Q$	T = 0 hold, T = 1 toggle

QUICK TRICK
JK is the best friend: J works like S (J = 1 can **set**), K works like R (K = 1 can **reset**), and J = K = 1 does the new thing: **toggle**. That is why JK has no invalid state.

QUICK TRICK
Quick answer for any JK question: read (J, K): **00 → same, 01 → 0, 10 → 1, 11 → opposite**. No need for the equation.

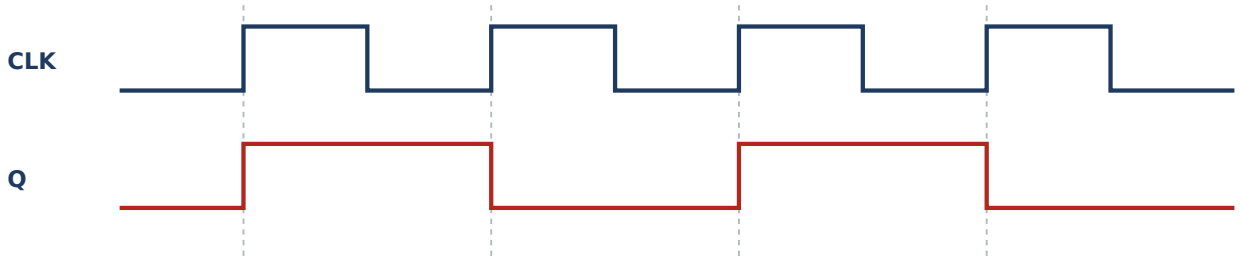
Timing diagrams

D flip-flop (rising-edge). At each rising edge (dashed line), Q copies the value that D had *just before* the edge. Changes of D between the edges are ignored.



QUICK TRICK
How to read a timing diagram: go to each active clock edge, look at the inputs **just before** the edge, and write the new Q. Q does not change anywhere else.

T flip-flop with T = 1. Q toggles at every rising edge, so Q has **half the frequency** of the clock. It is a **divide-by-2** circuit. (This is the idea of counters.)



Excitation tables — "what inputs do I need to get this change?"

An **excitation table** works backwards. You know the present state Q and the state you **want** (Q^+). It tells you which inputs to apply. X = don't care (0 or 1 both work). We need these in counter and state machine design.

Q	Q+	SR: S R	JK: J K	D	T
0	0	0 X	0 X	0	0
0	1	1 0	1 X	1	1
1	0	0 1	X 1	0	1
1	1	X 0	X 0	1	0

QUICK TRICK

Excitation tricks:
D: $D = Q^+$ (just copy what you want).
T: $T = Q \oplus Q^+$ (T = 1 only when the state must change).
JK: J decides about going **up** (from 0), K decides about going **down** (from 1). To go 0 → 1 you need J = 1 (K = X). To go 1 → 0 you need K = 1 (J = X). To stay at 0 you need J = 0 (K = X). To stay at 1 you need K = 0 (J = X).
SR: the same idea, but the other input must be 0 where JK has X for the going-up and going-down cases.

11.4 Race-around condition and master–slave flip-flop

● **LEVEL 3 · GATE**

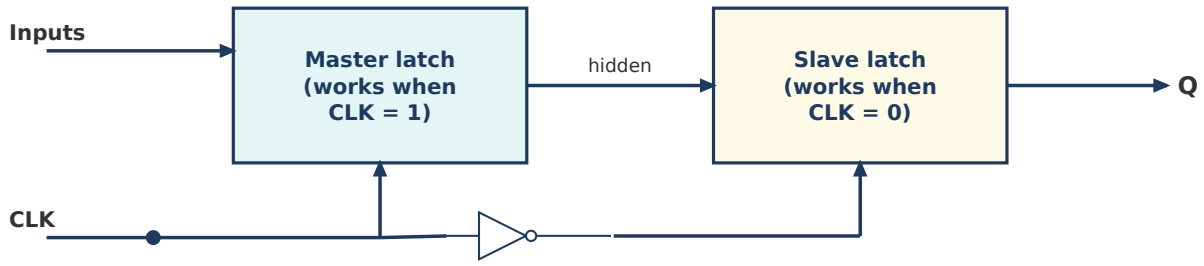
The problem. Take a **level-triggered** JK flip-flop with **J = K = 1**. While the clock stays HIGH, Q toggles. But the new Q is fed back and toggles it **again**, and again, as long as the clock is HIGH (each toggle takes one propagation delay). When the clock falls, the final Q is **unpredictable**. This is called the **race-around condition**.

GATE POINT

Race-around happens **only** when: (1) the flip-flop is **level triggered**, (2) **J = K = 1** (a T flip-flop with T = 1 also), and (3) the clock HIGH time is longer than the propagation delay.

Cures: (1) use **edge triggering**, (2) use a **master–slave** flip-flop, (3) make the clock pulse shorter than the propagation delay (hard to control).

Master–slave idea. Two latches in a row. The **master** works when CLK = 1. The **slave** works when CLK = 0 (it gets the inverted clock). So the output Q can change only when the clock **falls**, and the output cannot feed back to the master during the same clock level.



QUICK TRICK

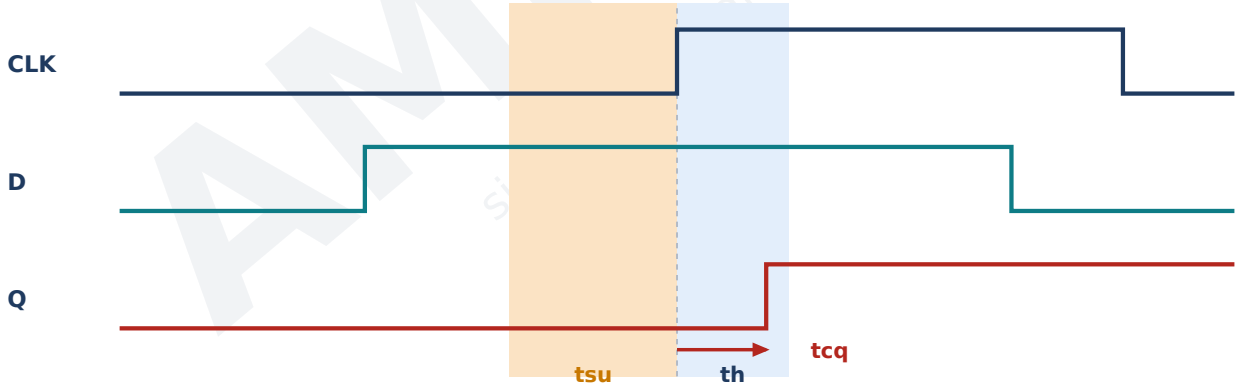
Master–slave = two doors in a corridor. Only one door is open at a time. So the signal can never run through both. That stops the race.

11.5 Setup time, hold time, clock-to-Q delay

● **LEVEL 3 · GATE**

A flip-flop needs its input to be **steady for a short time around the clock edge**.

Name	Symbol	Meaning
Setup time	t_{su}	The input must be **stable BEFORE** the active clock edge for at least this long.
Hold time	t_h	The input must **stay stable AFTER** the clock edge for at least this long.
Clock-to-Q delay	t_{cq}	Time after the clock edge until the output Q actually changes.



QUICK TRICK

Remember: Setup = BEFORE the edge. Hold = AFTER the edge. If D changes inside the shaded window, the flip-flop may store a wrong value (this is called metastability).

Maximum clock frequency. The signal leaves one flip-flop, goes through logic, and must reach the next flip-flop **before** the next edge:

```
T(clk) must be at least:    t_cq + t_logic(max) + t_su          f(max) = 1 / T(clk)
Hold check :                t_cq + t_logic(min)  >=  t_hold
```

Example (Level 3)

Given: t_{cq} = 2 ns, logic delay = 6 ns, t_{su} = 2 ns.

Concept: smallest clock period = t_{cq} + logic + t_{su}.

Calculation: 2 + 6 + 2 = 10 ns → f_{max} = 1 / 10 ns.

Answer: 100 MHz.

EXAM TRAP

Use the **maximum** logic delay for the frequency (setup) check. Use the **minimum** logic delay for the hold check. Changing the clock speed can fix a setup problem but **cannot** fix a hold problem.

Quick-Solve Tricks — Summary

Topic	Trick
SR	Set = 1, Reset = 0, 00 hold, 11 illegal. NOR latch invalid at 11, NAND latch invalid at 00.
JK	00 same, 01 → 0, 10 → 1, 11 → opposite (toggle).
D	Q^+ = D. Just copy D at the edge.
T	T = 1 toggles. Divide by 2.
Excitation	D = Q+. T = Q ⊕ Q+. JK: J decides going up, K decides going down.
Timing diagram	Look at the inputs just BEFORE each active edge.
Latch vs FF	Latch = level. Flip-flop = edge.
Race-around	Level-triggered JK with J = K = 1 and a long clock. Cure: master-slave or edge-triggering.
Setup / hold	Setup = before the edge. Hold = after the edge.
Max frequency	T(clk) ≥ t_cq + t_logic + t_su

GATE QUICK CHECK — Chapter 11

```
SR : Q+ = S + R'Q  (SR = 0)       JK : Q+ = JQ' + K'Q
D  : Q+ = D                        T  : Q+ = T xor Q
Excitation :  D = Q+      T = Q xor Q+
              JK: 0->1 (1,X)   1->0 (X,1)   0->0 (0,X)   1->1 (X,0)
Max clock frequency = 1 / (t_cq + t_logic + t_su)
```

Traps: NOR vs NAND latch invalid state · JK toggle · race-around needs a level-triggered JK with J = K = 1 · a T flip-flop with T = 1 halves the frequency · hold violation is not cured by a slower clock.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. A JK flip-flop has J = K = 1 and Q = 0. What is Q after the clock edge?

Q2. What happens in an SR flip-flop if S = R = 1?

Q3. What is the main difference between a latch and a flip-flop?

● **LEVEL 2 · EXAM**

Q4. A T flip-flop with T = 1 is clocked at 2 MHz. What is the frequency of Q?

Q5. A JK flip-flop starts with Q = 0. (J, K) at four successive clock edges are (1,0), (1,1), (0,1), (1,1). Find Q after each edge.

Q6. A D flip-flop starts with Q = 0. D = 1, 0, 1, 1 just before four successive active edges. Find Q after each edge.

● **LEVEL 3 · GATE**

Q7. Under which three conditions does race-around occur?

Q8. t_{cq} = 3 ns, logic delay = 9 ns, t_{su} = 3 ns. Find the maximum clock frequency.

Q9. NAND SR latch: what happens when S' = 0 and R' = 0?

Q10. Three T flip-flops (T = 1) are connected in a chain, each output driving the next clock. The first clock is 16 MHz. Find the frequency at the last output and the number of states.

Q11. Which inputs are needed for a JK flip-flop to change from 1 to 0? From 0 to 1?

Q12. A flip-flop has t_h = 4 ns. The previous flip-flop has t_{cq} = 2 ns and the logic between them can be as fast as 1 ns. Is there a hold violation?

Answers

Q	Solution
1	J = K = 1 → toggle. Q goes from 0 to **1**.
2	**Invalid** (illegal) input. The outputs are not reliable.
3	A latch is **level**-triggered. A flip-flop is **edge**-triggered (it changes only at the clock edge).
4	T = 1 toggles at each edge, so the output frequency is half: **1 MHz**.
5	(1,0) set → **1**. (1,1) toggle → **0**. (0,1) reset → **0**. (1,1) toggle → **1**. Sequence: **1, 0, 0, 1**.
6	Q follows D at each edge: **1, 0, 1, 1**.
7	(1) Level-triggered JK flip-flop, (2) J = K = 1, (3) clock HIGH time longer than the propagation delay.
8	**Given:** 3, 9, 3 ns. **Concept:** T = t_cq + logic + t_su. **Calculation:** 3 + 9 + 3 = 15 ns → 1/15 ns. **Answer: 66.7 MHz.**
9	**Invalid** condition. Both outputs Q and Q' become 1, and the final state is unpredictable when the inputs go back to 1.
10	Each stage divides by 2: 16 → 8 → 4 → 2. Output = **2 MHz** (divide by 2^3 = 8). The chain has 2^3 = **8 states**.
11	1 → 0: **K = 1** (J is X). 0 → 1: **J = 1** (K is X).
12	Hold check: t_cq + logic(min) = 2 + 1 = 3 ns, which is less than t_h = 4 ns → **yes, hold violation**. (Slowing the clock does not help; the logic delay or clock skew must be changed.)

Next: Chapter 12 — Flip-Flop Conversion (SR → JK, JK → T, JK → D, D → T, T → D… step by step with the excitation table).